

Bharat Badgujar

Linux module 2

Assignment 5

1.

Enable and configure SSH on your Linux server so that only key-based authentication is allowed and password authentication is disabled.

Ans : Configure SSH (Key Authentication Only)

Edit SSH config:

```
sudo nano /etc/ssh/sshd_config
```

Change:

```
PubkeyAuthentication yes
```

```
PasswordAuthentication no
```

```
PermitRootLogin no
```

Restart SSH:

```
sudo systemctl restart ssh
```

2.

Set up a basic firewall using UFW or firewalld to allow only SSH (port 22) and HTTP (port 80) traffic, and block all other incoming connections.

Hint: After applying rules, use a port scanning tool like nmap from another machine to verify only the allowed ports are open.

Ans : Configure UFW Firewall

```
sudo apt install ufw -y
```

```
sudo ufw allow 22/tcp
```

```
sudo ufw allow 80/tcp
```

```
sudo ufw default deny incoming
```

```
sudo ufw enable
```

```
sudo ufw status
```

Verify from another machine:

```
nmap <server_ip>
```

3.

Write a shell script `automate_backup.sh` that creates a backup of your `/home` directory and saves it with a timestamp in `/backups`.

Ans : **automate_backup.sh**

```
#!/bin/bash
```

```
DATE=$(date +%F_%H-%M-%S)
```

```
mkdir -p /backups
```

```
tar -czf /backups/home_backup_$(DATE).tar.gz /home
```

```
echo "Backup Completed"
```

Make executable:

```
chmod +x automate_backup.sh
```

4.

Check the current status of SELinux on your server, switch it to enforcing mode if it is not already, and verify the change using appropriate commands.

Hint: Use the commands `getenforce` and `setenforce`.

Ans : **Check SELinux**

```
getenforce
```

```
sudo setenforce 1
```

```
getenforce
```

Expected output:

Enforcing

Kali Linux: SELinux is usually not available (uses AppArmor).

5.

Use ChatGPT or Copilot to generate a checklist of at least 5 best practices for securing a Linux server, and compare it with your own server configuration to identify any missing steps.

Ans : **Linux Server Security Checklist**

- ☒ Use SSH keys

- ☒ Disable password login
- ☒ Enable firewall
- ☒ Take regular backups
- ☒ Keep the system updated